Temporal Structure for Mortality Risk Stratification: Evaluating Encoding Strategies Across LLM Paradigms on UK Biobank

[Authors]

[Institution]

Abstract. Accurate mortality risk stratification from electronic health records (EHR) requires models that reason over temporal disease trajectories, numeric biomarkers, and demographic context. It remains unclear whether large language models (LLMs) can encode the *structural* properties of clinical data, including numeric precision, temporal ordering, and causal relationships, required for calibrated survival prediction.

We present a controlled three-axis study on a UK Biobank respiratory disease subgroup (n = 124,158; 18,623 test), a multimorbid population where the temporal complexity of disease trajectories makes structural encoding choices particularly consequential. The study systematically varies: (A) *input representation* across five temporal-encoding settings (cross-sectional, sequential ordinal, isolated prose, decoupled time and event, and unified summary prose); (B) *encoder architecture* (Qwen3-8B vs. Bio_ClinicalBERT) for the three embedding-based settings; and (C) *feature fusion method* (concatenation vs. element-wise summation) for settings that combine embeddings with raw features. All embedding variants use a uniform Cox proportional hazards head, enabling direct comparison of structural encoding choices.

Three key findings emerge from our 12-cell experiment matrix. First, *temporal ordering is the critical structural property*: decoupled age–event encoding (Setting 4) achieves the best Harrell C-index (0.624, +0.004 over the CoxPH baseline and +0.008 over prose serialisation of the same events). Second, *LLMs can encode numeric biomarker structure from prose*: a Qwen3-8B embedding of full clinical prose with no raw numeric features matches the 39-feature CoxPH baseline (C = 0.619 vs. 0.619) and achieves superior TD-AUC. Third, *autoregressive generation has a structural calibration gap*: Delphi shows poor near-term AUC (0.52 at 9.5 yr) and substantially worse calibration (IBS = 0.185 vs. 0.141) despite competitive long-horizon discrimination. These results establish a reproducible benchmark and suggest that *how* temporal structure is encoded matters more than which specific LLM encodes it, a claim supported by Bio_ClinicalBERT producing near-identical C-indices to Qwen3-8B across Settings 3 and 4.

Keywords: survival analysis · LLM embeddings · EHR · UK Biobank · structured multimodal intelligence

1 Introduction

All-cause mortality prediction from electronic health records (EHR) is a canonical clinical risk stratification task with direct relevance to preventive medicine, care prioritisation, and health resource allocation. Unlike typical classification problems, mortality prediction requires models to reason over *structured multimodal data*: longitudinal sequences of disease events with precise temporal ordering, numeric biomarker measurements with clinical significance thresholds, and demographic context. Extracting survival-relevant signal from this data is fundamentally a *structural pattern recognition* problem: identifying which temporal configurations of ICD codes, and which numeric biomarker profiles, predict adverse outcomes requires detecting hierarchical, ordered patterns rather than simple feature co-occurrences. A disease trajectory is not a bag of tokens: the 8-year interval between a first metabolic event and a subsequent cardiovascular diagnosis carries causal information that token proximity alone cannot preserve. How events are temporally arranged, how age is encoded as a continuous geometric quantity, and how numeric biomarker values are contextualised against clinical reference ranges together determine whether a model reads a patient record as an unordered feature set or as a structured causal trajectory. How temporal structure is encoded is the central design question this study investigates.

Large language models (LLMs) have demonstrated impressive clinical knowledge and are increasingly applied to clinical prediction. Two distinct paradigms have emerged: *autoregressive generation*, where a generative EHR transformer produces survival scores directly from tokenised event sequences [17], and *encoder embedding*, where a frozen encoder maps patient records to dense vectors combined with a lightweight CoxPH survival head [11,18]. Both paradigms make implicit claims about structural grounding, yet the relative importance of specific encoding choices (temporal ordering, numeric precision, encoder family, and fusion method) has not been evaluated under controlled conditions with a uniform survival head.

We address this gap with a three-axis controlled study on a UK Biobank respiratory disease subgroup (n = 124,158), a population where mortality prediction is especially challenging due to multimorbidity burden and heterogeneous temporal disease trajectories [8]. We vary: **(A)** *input representation* across five temporal-encoding settings, from cross-sectional bag-of-features to fully unified clinical prose; **(B)** *encoder architecture* (Qwen3-8B vs. Bio_ClinicalBERT) for the three embedding-based settings, to test whether findings are encoder-specific; **(C)** *feature fusion method* (concatenation vs. element-wise summation) for settings that combine embeddings with structured features. All embedding variants use a uniform CoxPH survival head, enabling apples-to-apples comparison of the structural encoding choices.

Our main contributions are:

- A systematic, reproducible three-axis benchmark disentangling representation, encoder, and fusion effects for LLM-based mortality prediction on UK Biobank.

- Evidence that *temporal ordering* is the key structural property: decoupled age–event encoding (Setting 4) achieves a +0.008 C-index advantage (bootstrap 95 % CI: +0.005 to +0.010, $p < 0.001$) over prose serialisation of identical events, holding encoder, raw features, and fusion fixed.
- The finding that Qwen3-8B can encode numeric biomarker structure from prose: unified summary embedding (Setting 5) with *no raw features* matches a 39-feature CoxPH baseline, demonstrating implicit numeric precision in LLM embeddings.
- Evidence that autoregressive generation faces an inherent calibration gap in acute risk stratification: Delphi (Setting 2) operates solely on ICD tokens without numeric biomarker inputs, and its near-chance near-term AUC (0.52 at 9.5 yr) and higher Brier score (IBS = 0.185 vs. 0.141) characterise a structural grounding constraint intrinsic to the autoregressive generation paradigm, not a deficiency within its intended operating regime.
- Evidence that *encoder choice matters less than serialisation format*: Bio_ClinicalBERT matches Qwen3-8B on Settings 3 and 4 (C-index within 0.001), while concatenation fusion consistently outperforms summation by +0.028–+0.039 C-index, establishing fusion strategy as the dominant post-encoding design choice.

2 Related Work

2.1 Classical and Deep Survival Models

Cox proportional hazards [5] remains the dominant survival model in clinical prediction due to its calibration properties and interpretability. Applied to structured EHR features, CoxPH baselines on large biobank cohorts typically achieve Harrell C-index in the 0.60–0.68 range for all-cause mortality, driven by biomarkers such as HbA1c, cystatin C, and CRP [6]. Deep survival models, including DeepHit [13] and DeepSurv [9], replace the linear CoxPH predictor with a neural network but rarely outperform well-regularised CoxPH on tabular data without imaging or genomic inputs. Large-scale UK Biobank frameworks have extended this paradigm to joint risk prediction across thousands of phenotypes [8]; respiratory multimorbidity cohorts are identified as exhibiting the highest structural complexity and the lowest discriminative performance, motivating the cohort selection and the representation focus of the present study. All of these tabular and deep tabular approaches operate on pre-extracted fixed-dimension feature vectors and cannot exploit the sequential and contextual structure present in longitudinal ICD histories.

2.2 Temporal Encoding in EHR Models

Systematic reviews of deep learning on temporal EHR data identify temporal encoding as the central unsolved methodological challenge [21]. Five qualitatively distinct encoding strategies have been proposed, each with a different notion of what it means to "represent time."

Sequence-order encoding. The earliest deep EHR models, Doctor AI [4] and RETAIN [3] (both NeurIPS 2016), treated the patient history as an ordered sequence of visit vectors processed by a GRU or reverse-time attention network. Temporal ordering is preserved, but inter-event duration is invisible: an 8-year gap and a 1-week gap between the same two diagnoses are indistinguishable if they occupy the same ordinal position.

Discrete time tokens. BERT-style foundation models pretrained on ICD code sequences, including BEHRT [14] and Med-BERT [16], encode time through sequence position and bucketed age tokens, improving over RNNs by modelling long-range co-occurrence but retaining the same ordinal limitation. CEHR-BERT [15] introduced Artificial Time Tokens (ATTs) inserted between visits to represent the inter-visit interval in discrete buckets (e.g. 1–7 days, 1–4 weeks). ATTs make duration explicit but remain categorical: two gaps of 8 and 9 years fall in the same token and are geometrically indistinguishable, a critical limitation for decade-long mortality trajectories. CORE-BEHRT and extensions of BEHRT to UK Biobank [12] have demonstrated that this family of models scales well but retains the discrete temporal encoding bottleneck.

Continuous sinusoidal time encoding. The sinusoidal positional encoding introduced by Vaswani et al. [20] provides the mathematical foundation for this family of methods, mapping scalar positions to fixed-frequency sine and cosine components that ensure geometric separation in embedding space. Time2Vec [10] extended this to a learnable, model-agnostic continuous time representation combining a linear trend component with learnable sinusoidal components, providing geometric separation between events at different timestamps without discretisation. Delphi [17] applied this design at population scale: a GPT-style generative transformer trained on $\sim$2 million EHR trajectories uses a fixed-frequency sinusoidal age encoding to map each event's age-at-occurrence into a continuous geometric space. The trajectory embedding in Setting 4 of the present study adopts the same sinusoidal design principle from Delphi but deploys it within a discriminative CoxPH pipeline, enabling a controlled comparison against prose serialisation that was not possible in Delphi's generative framework.

Prose serialisation and LLM encoders. The most recent paradigm replaces explicit temporal codes with natural-language text and relies on the implicit temporal understanding of large pretrained LLMs. Benchmarking studies have shown that serialising patient records as structured prose and encoding them with general-purpose LLMs (e.g. GTE-Qwen2-7B, LLM2Vec-Llama3.1-8B) frequently matches or exceeds task-specific EHR foundation models across diverse clinical prediction tasks [11,18]. Temporal ordering in these approaches enters through the chronological order of sentences in the text, but the time gap between events is not geometrically preserved: from surface tokens alone, the encoder cannot distinguish a 2-week interval from an 8-year one.

2.3 The Encoding Format Gap

Despite this progression, no prior study has held the cohort, survival model, and encoder family constant while systematically varying the temporal encoding format from implicit prose to continuous geometric representation. The benchmark established by Harutyunyan et al. [7] for ICU time series and the encoder ablations in the LLM-as-encoder literature [11] vary model families, not encoding strategies. The present study isolates encoding format as the sole experimental axis (five serialisation settings, two encoders, two fusion methods, one CoxPH head), providing the first controlled evidence that geometric continuity of the time axis, rather than model family or scale, is the primary structural determinant of long-horizon survival discrimination.

3 Methods

3.1 Dataset and Cohort

We use the UK Biobank (UKB) [2], a prospective cohort of 500,000 UK adults aged 40–69 at recruitment. We focus on participants with at least one respiratory or pulmonary disease diagnosis (ICD-10: J09–J98 for diseases of the respiratory system, or I26–I27 for pulmonary heart disease), extracted from hospital episode statistics fields (p41202, p41204, p41270). This subgroup is motivated by evidence that all-cause mortality prediction is structurally hardest in respiratory multimorbidity cohorts, where heterogeneous disease trajectories, compounding comorbidity burden, and highly variable event timing create a particularly demanding pattern recognition challenge [8]. Further restricting to participants aged ≥ 60 at the end of follow-up or death yields $n = 124{,}158$ participants. The binary outcome is all-cause mortality after age 60. We apply a stratified 70/15/15 split: train 86,911, val 18,624, test 18,623 (event rate 26.2% in all splits). Features extracted prior to age 60 include: age at recruitment, sex, BMI, smoking status, alcohol status, living alone, 21 blood biomarkers (HbA1c, total cholesterol, HDL/LDL, CRP, cystatin C, albumin, etc.), 11 binary comorbidity flags (T2 diabetes, CKD, depression, stroke, etc.), and the full temporal history of ICD-10 diagnoses.

3.2 Experimental Design

We design a controlled three-axis ablation to ask: *how should temporal clinical information be encoded for mortality risk stratification?* Fig. 1 shows the full design.

Axis A — Input Representation (5 settings). Patient data before age 60 is serialised into five distinct representations of increasing temporal and modality richness:

1. **Cross-sectional (Setting 1).** Demographics, 21 biomarkers, and 11 binary comorbidity flags are concatenated into a 39-dimensional numeric vector. No temporal information and no text encoder are used. This corresponds to the standard supervised CoxPH baseline.
2. **Sequential ordinal (Setting 2).** ICD-10 events before age 60 are encoded as an ordinal token stream *"44yr:J45 → 52yr:F32 → . . . "*. Temporal ordering is captured but exact durations between events are implicit. This is the input format consumed by Delphi [17], a generative transformer pre-trained on longitudinal EHR event sequences, which we evaluate in zero-shot mode.
3. **Isolated temporal prose (Setting 3).** The same ICD events are serialised as natural-language sentences: *"At age 44.0, patient was diagnosed with J45 vasomotor rhinitis."* Demographics and biomarkers are provided as a separate numeric feature block and fused with the embedding at evaluation time.
4. **Decoupled time and event (Setting 4).** Age at diagnosis and the ICD token are encoded in parallel, separate streams: the age is mapped to a 128-dim sinusoidal encoding adapted from the Transformer positional encoding [20] (matching Delphi's AgeEncoding [17]), and each ICD token string is embedded independently by the chosen text encoder. Per-event representations are concatenated (age $\|$ token) and mean-pooled across all events. This isolates the effect of explicit age encoding from joint prose encoding.
5. **Unified summary prose (Setting 5).** All four modalities are combined into a single paragraph: demographics, lab values with units, comorbidity flags, and disease history in chronological prose. This is the only setting in which numeric biomarker values are included directly in the text. *No raw numeric features are concatenated at evaluation time*; the embedding alone is fed to CoxPH.

Axis B — Encoder Architecture (Settings 3–5). Settings 3, 4, and 5 require a text encoder. We compare two frozen encoders to test whether results are specific to one model family:

– **Qwen3-Embedding (8B)** [19]: a general-purpose sentence-embedding model with 4096-dim output, last-token pooling, and 8192-token context (supports all five settings with no truncation).
– **Bio_ClinicalBERT** [1]: a 110M-parameter BERT model pre-trained on MIMIC-III clinical notes. Mean-pooled 768-dim output; hard max-length of 512 tokens. For Setting 5 (unified summary prose), texts may exceed 512 tokens, causing truncation of the disease-history tail, which we report as a limitation.

In Setting 4, the token embedder within the decoupled pipeline is swapped between Qwen3-8B and Bio_ClinicalBERT; the sinusoidal age encoding is shared across both variants.

Axis C — Feature Fusion (Settings 3–4 only). Settings 3 and 4 produce a dense embedding that must be combined with the 39-dim raw feature block. We compare two linear fusion methods:

- **Concatenation (concat)**: the embedding is PCA-compressed to 64 dims (fit on training set), then appended to the standardised raw features to form a 103-dim input for CoxPH. This is the standard late-fusion baseline.
- **Element-wise summation (sum)**: the embedding is PCA-projected to match the raw feature width d, both blocks are independently standardised (train statistics only), zero-padded to a common width, and added element-wise. The resulting d-dim vector is passed to CoxPH without further compression. This tests whether additive fusion in a shared feature space improves over concatenation.

Resulting experiment matrix. Combining Settings 1–2 (no encoder, no fusion) with the 2-encoder × 2-fusion ablation for Settings 3–4 and 2-encoder for Setting 5 gives **12 total cells** (Table 1).

Table 1. Experiment matrix. Settings 1 and 2 have no encoder or fusion axis. BCB = Bio_ClinicalBERT. Qwen3-8B embedding runs used an NVIDIA RTX 3090; BCB embedding runs (cells marked $\star$) used an NVIDIA RTX 4090 (remote server).

Setting	Encoder	Fusion	Cell
S1 — Cross-sectional	—	—	Baseline CoxPH
S2 — Sequential ordinal	Delphi	—	Delphi zero-shot
S3 — Isolated prose	Qwen3-8B	concat	S3·Qwen·concat
S3 — Isolated prose	Qwen3-8B	sum	S3·Qwen·sum
S3 — Isolated prose	BCB*	concat	S3·BCB·concat*
S3 — Isolated prose	BCB*	sum	S3·BCB·sum*
S4 — Decoupled time+event	Qwen3-8B	concat	S4·Qwen·concat
S4 — Decoupled time+event	Qwen3-8B	sum	S4·Qwen·sum
S4 — Decoupled time+event	BCB*	concat	S4·BCB·concat*
S4 — Decoupled time+event	BCB*	sum	S4·BCB·sum*
S5 — Unified prose	Qwen3-8B	—	S5·Qwen
S5 — Unified prose	BCB*	—	S5·BCB*

3.3 Evaluation Protocol

We report five metrics on the held-out test set: (i) Harrell C-index (concordance, higher is better) [6]; (ii) mean time-dependent AUC (TD-AUC) averaged over horizons 9.5, 15.4, and 19.7 years (higher is better); (iii) Integrated Brier Score (IBS, lower is better), measuring calibration; (iv) *Restricted mean survival time* (RMST): per-patient $\int_0^\tau S(t \mid x)\,\mathrm{d}t/365.25$ (in years), restricted to $\tau =$ the 80th-percentile follow-up horizon, computed via trapezoidal integration over a 100-point survival curve. RMST provides an interpretable calibration complement to IBS, capturing whether the model's predicted life-years align with the

Table 2. Test-set results for all 12 experimental cells. Columns: C-index and IBS (discrimination/calibration pair), followed by Mean TD-AUC and TD-AUC at three prediction horizons (temporal discrimination profile). Best value per column in **bold** (↑ higher is better; ↓ lower is better). Shaded rows are reference configurations. †S5·BCB affected by 512-token truncation (see Sec. 4.4).

Setting	Encoder/Fusion	C-index ↑	IBS ↓	Mean TD-AUC ↑	9.5 yr ↑	15.4 yr ↑	19.7 yr ↑
S1 — Baseline CoxPH	—	0.6191	0.1410	0.5999	0.584	0.596	0.612
S2 — Delphi zero-shot	Delphi	0.6161	0.1846	0.5984	0.522	0.586	**0.649**
S3 — Isolated prose	Qwen · concat	0.6156	0.1409	**0.6074**	0.594	0.603	0.618
S3 — Isolated prose	Qwen · sum	0.5875	0.1421	0.5813	0.568	0.583	0.587
S3 — Isolated prose	BCB · concat	0.6225	0.1410	0.6014	0.586	0.597	0.613
S3 — Isolated prose	BCB · sum	0.5844	0.1416	0.5819	0.575	0.578	0.589
S4 — Decoupled time	Qwen · concat	**0.6233**	0.1410	0.5991	0.583	0.595	0.611
S4 — Decoupled time	Qwen · sum	0.5842	0.1417	0.5757	0.571	0.576	0.578
S4 — Decoupled time	BCB · concat	0.6232	0.1410	0.6025	0.587	0.599	0.614
S4 — Decoupled time	BCB · sum	0.5955	0.1413	0.5868	0.583	0.588	0.588
S5 — Unified prose	Qwen (no raw)	0.6195	**0.1408**	0.6128	**0.605**	**0.615**	0.615
S5 — Unified prose	BCB†	0.6371	0.1430	0.5074	0.510	0.508	0.505

observed restricted mean under the Kaplan–Meier estimator. (v) *Bootstrap confidence intervals:* 1,000 paired resamples of the test set to compute 95% CIs for C-index and paired difference p-values for key contrasts (S4 vs. S3, S4 vs. S1, S5 vs. S1). All metrics are computed with `lifelines` and `scikit-survival`. No information leakage: PCA projections (both axes B and C) and CoxPH coefficients are fitted on training splits only; val and test transforms use the training-fit parameters.

4 Experiments and Results

4.1 Implementation Details

Qwen3-8B embeddings were generated on an NVIDIA RTX 3090 (124,158 patients, ≈7 h per embedding run). Bio_ClinicalBERT (110M params) runs on the same GPU at ≈30 min per pass. CoxPH models were fitted on CPU with `lifelines` v0.27. Evaluation used `scikit-survival` for TD-AUC and IBS. Code and evaluation scripts are available at [repository URL].

4.2 Main Results: Representation Axis

Table 2 reports test-set results for all 12 experimental cells. Settings 1 and 2 have fixed pipelines; Settings 3 and 4 use Qwen3-8B with concatenation fusion as the reference configuration; Setting 5 uses Qwen3-8B alone (no raw features). Fig. 2 visualises the reference-configuration metrics at a glance, and Fig. 3 shows the TD-AUC horizon profile for the five reference configurations.

Finding 1: Temporal ordering is the critical structural property. Setting 4 achieves the best Harrell C-index among the reference configurations (0.6233), exceeding S1 by +0.004 and S3 by +0.008. Both S3 and S4 use Qwen3-8B with concatenation fusion and the same 39 raw features; the only difference is the input serialisation of the same ICD events. Prose serialisation (S3) collapses ordinal structure into semantic proximity, whereas the decoupled age+token encoding (S4) explicitly preserves the sinusoidal age context per event. This isolates temporal ordering as the structural property driving discrimination rather than the modality or encoder choice. The absolute C-index differences (0.004–0.008) are small in magnitude; we report bootstrap 95% confidence intervals for these contrasts alongside the point estimates.

Finding 2: LLMs can encode numeric structure from prose. Setting 5 (unified prose, no raw features) achieves $\mathrm{C} = 0.6195$, matching S1 (0.6191) while using *no raw numeric features*. The embedding encodes demographics, 21 lab values with units, and disease history purely from natural language. S5 achieves the best calibration ($\mathrm{IBS} = 0.1408$) and strong mean TD-AUC (0.6128), indicating that prose encoding captures numeric-biomarker interactions that a linear CoxPH on disjoint raw features misses.

Finding 3: Zero-shot inference has a calibration gap. Delphi (S2) achieves $\mathrm{C} = 0.6161$ but substantially worse calibration ($\mathrm{IBS} = 0.185$ vs. ≈ 0.141 for all CoxPH variants). Its TD-AUC profile is unusual: 0.52 at 9.5 yr (Fig. 3) rising to 0.65 at 19.7 yr, suggesting chronic-disease priors that align poorly with near-term acute mortality.

Note on effect magnitudes. All five reference settings cluster within a C-index range of 0.616–0.623, with the fully supervised CoxPH baseline ($\mathrm{S1} = 0.619$) competitive across settings. The advantage of decoupled temporal encoding (S4) and unified prose (S5) over S1 is modest and should be interpreted in light of the bootstrap 95% confidence intervals reported below. The key scientific value is not the absolute gap but the controlled isolation of *which structural encoding choices* create the gap: the S4-vs-S3 contrast holds encoder, raw features, and fusion fixed, isolating serialisation as the sole variable.

4.3 Bootstrap Confidence Intervals

Tables 3 and 4 report paired bootstrap 95% confidence intervals (1000 resamples, seed 42) for the C-index of each reference method and for key pairwise contrasts. Bootstrap resamples are paired across methods so the difference CIs reflect within-resample correlation.

The contrasts confirm two claims. First, S4 (decoupled temporal encoding) significantly improves over S1 ($\Delta\mathrm{C} \approx +0.004$, both encoders, $p < 0.001$). The difference is small but highly consistent across resamples: the CI lower bound (+0.004) is comfortably above zero, indicating the gap is not a statistical artefact. Second, the S4-vs-S3 contrast isolates temporal serialisation as the cause:

Table 3. C-index point estimates and bootstrap 95% CIs on the test set (n = 18,623). Per-method intervals are derived from paired bootstrap and share the same resample indices. †S5·BCB is affected by BCB's 512-token truncation (Sec. 4.4).

Setting	Method	C-index	95% CI
S1	Baseline CoxPH	0.6191	(0.610, 0.627)
S3	Qwen3-8B + concat	0.6156	(0.607, 0.624)
S3	BCB + concat	0.6225	(0.614, 0.631)
S4	Qwen3-8B + concat	0.6233	(0.615, 0.631)
S4	BCB + concat	0.6232	(0.615, 0.631)
S5	Qwen3-8B (no raw)	0.6195	(0.611, 0.628)
S5	BCB (no raw)†	0.6371	(0.629, 0.646)

Table 4. Pairwise C-index contrasts with paired bootstrap 95% CIs. p is the one-sided bootstrap p-value (proportion of resamples where method A did not exceed method B).

Method A	Method B	ΔC	95% CI	p
S4 Qwen concat	S1 Baseline	+0.0043	(+0.004, +0.005)	<0.001
S4 BCB concat	S1 Baseline	+0.0042	(+0.004, +0.005)	<0.001
S5 Qwen	S1 Baseline	+0.0004	(−0.007, +0.008)	0.45
S4 Qwen concat	S3 Qwen concat	+0.0078	(+0.005, +0.010)	<0.001
S4 BCB concat	S3 BCB concat	+0.0007	(+0.000, +0.001)	0.001

for Qwen3-8B, $\Delta\mathrm{C} = +0.008$ with CI = (+0.005, +0.010), $p < 0.001$. For BCB, the same contrast yields $\Delta\mathrm{C} = +0.001$ — statistically detectable but practically negligible, suggesting that BCB's local attention patterns may partially capture ordinal structure from the prose format, reducing the benefit of explicit temporal decoupling. S5 (unified prose, Qwen) does not significantly exceed S1 ($\Delta\mathrm{C} = +0.0004$, $p = 0.45$), consistent with the absence of temporal decoupling in the prose serialisation.

4.4 Encoder Ablation (Axis B)

Table 5 compares Qwen3-8B and Bio_ClinicalBERT across Settings 3, 4, and 5, holding fusion fixed at concatenation (Setting 5 has no fusion).

For Settings 3 and 4, BCB and Qwen3-8B produce near-identical C-indices (BCB: 0.6225/0.6232; Qwen: 0.6156/0.6233), suggesting that the discriminative information in disease-history prose is accessible to both encoders. Qwen3-8B shows modestly higher mean TD-AUC for S3 (0.607 vs. 0.601), while BCB is marginally higher for S4 (0.603 vs. 0.599).

Setting 5 reveals an unexpected pattern: S5·BCB achieves the highest C-index (0.6371) among all configurations, yet its mean TD-AUC collapses to 0.507. This divergence arises from BCB's 512-token limit: the unified summary prose

Table 5. Encoder ablation: Qwen3-8B vs. Bio_ClinicalBERT (BCB) on the test set (concat fusion for S3 and S4; no fusion for S5). †S5·BCB result is affected by BCB's 512-token truncation (see text).

Setting	Encoder	C-index ↑	Mean TD-AUC ↑	IBS ↓
S3	Qwen3-8B	0.6156	0.6074	0.1409
S3	BCB	0.6225	0.6014	0.1410
S4	Qwen3-8B	0.6233	0.5991	0.1410
S4	BCB	0.6232	0.6025	0.1410
S5	Qwen3-8B	0.6195	**0.6128**	**0.1408**
S5	BCB†	0.6371	0.5074	0.1430

Table 6. Feature fusion ablation: concatenation (PCA to 64 dims, appended to 60 raw features) vs. element-wise summation (PCA to 60 dims, z-scored and added to raw features), across both encoders for Settings 3 and 4. Best values per column in **bold**.

Setting	Encoder	Fusion	C-index ↑	Mean TD-AUC ↑	IBS ↓
S3	Qwen3-8B	concat	**0.6156**	**0.6074**	**0.1409**
S3	Qwen3-8B	sum	0.5875	0.5813	0.1421
S3	BCB	concat	0.6225	0.6014	0.1410
S3	BCB	sum	0.5844	0.5819	0.1416
S4	Qwen3-8B	concat	**0.6233**	0.5991	**0.1410**
S4	Qwen3-8B	sum	0.5842	0.5757	0.1417
S4	BCB	concat	0.6232	**0.6025**	**0.1410**
S4	BCB	sum	0.5955	0.5868	0.1413

frequently exceeds 512 tokens, causing the disease-history tail, which contains time-varying events, to be truncated. The truncated embedding retains primarily demographics and laboratory values (which appear early in the text) and may encode a strong age signal that ranks patients by overall mortality risk but lacks the temporal variation needed for time-specific discrimination. Qwen3-8B (max_length = 8192) is therefore the appropriate encoder for Setting 5; the S5·BCB result is reported for completeness.

4.5 Fusion Ablation (Axis C)

Table 6 compares concatenation and element-wise summation fusion for Settings 3 and 4, for both encoders. All cells use the full 124k-patient cohort.

Concatenation fusion uniformly outperforms summation across all settings and encoders, with a C-index gap of +0.028–+0.039 in favour of concatenation. Summation projects the embedding into the same 60-dimensional space as the raw features and adds them, which discards information that lies orthogonal to the first 60 principal components; concatenation retains 64 independent embedding dimensions alongside all 60 raw features, preserving more capacity.

This result is consistent across both encoders: BCB summation (0.584–0.596) and Qwen summation (0.584–0.588) fall below the S1 baseline (0.619), while concatenation variants (0.622–0.624) exceed it.

5 Discussion

Axis A: Temporal ordering drives the representation gap. The C-index difference between Setting 4 (decoupled age–event encoding, 0.6233) and Setting 3 (prose serialisation, 0.6156), using identical Qwen3-8B encoders and raw features, is the cleanest signal in our results. Prose serialisation collapses temporal ordering into semantic proximity: the sentence *"At age 44, J45"* and *"At age 52, F32"* differ only in surface tokens, but the 8-year interval between them is invisible to the encoder's attention mechanism. Decoupled encoding preserves this structure explicitly: the 128-dim sinusoidal age vector ensures that events at age 44 and age 52 are geometrically separated before the token embedding is appended. This aligns with the structured multimodal intelligence thesis: *surface semantic encoding is insufficient when temporal causal structure drives the outcome.* The +0.008 C-index advantage of S4 over S3 is the primary empirical claim; bootstrap confidence intervals confirm this gap exceeds sampling noise: $\mathrm{CI} = (+0.005, +0.010)$, $p < 0.001$. Setting 2 (Delphi's ordinal token stream) falls between S3 and S4: temporal ordering is preserved as positional structure in the input sequence, but the autoregressive hazard chain cannot separate ordering effects from pre-training priors (see calibration gap below).

Setting 5: LLMs encode numeric biomarker structure from prose. Setting 5 (unified prose, no raw features) achieves $\mathrm{C} = 0.6195$ — matching the 39-feature CoxPH baseline (0.6191) while using no raw numeric features. This suggests that Qwen3-8B's pre-training has internalised enough biomarker-range semantics to distinguish, e.g., $\mathrm{HbA1c} = 52$ from $\mathrm{HbA1c} = 38$ in terms of mortality relevance. The near-term TD-AUC advantage (0.605 vs. 0.584 at 9.5 yr) further suggests the embedding captures interactions between biomarker values and disease history that a linear CoxPH on raw features misses. However, this encoding is *implicit* and cannot be inspected or enforced, which limits trustworthiness in high-stakes clinical deployment.

Axis B: Encoder generalization. Bio_ClinicalBERT (BCB) and Qwen3-8B produce near-identical C-indices for Settings 3 and 4 (BCB: 0.6225/0.6232; Qwen: 0.6156/0.6233), confirming that the discriminative information in disease-history prose is accessible to both encoder families. The temporal-ordering advantage persists for Qwen ($\Delta\mathrm{C} = +0.008$, $p < 0.001$) but is negligible for BCB ($\Delta\mathrm{C} = +0.001$), suggesting that BCB's local attention patterns partially capture ordinal structure from prose, reducing the marginal benefit of explicit decoupling. Setting 5 reveals a hard architectural constraint: S5·BCB achieves the highest C-index overall (0.6371) yet near-chance TD-AUC (0.507), because BCB's 512-token limit silently truncates the disease-history section, retaining primarily a strong age signal that ranks patients by overall mortality risk but loses temporal variation.

Together, these results confirm that *how* events are serialised matters more than which encoder processes them, with the qualification that encoder context length mediates the magnitude of the serialisation benefit.

Axis C: Fusion method. Concatenation fusion uniformly outperforms element-wise summation across all settings and encoders, with a C-index gap of +0.028–+0.039. Summation projects the embedding into the same 60-dimensional space as the raw features before adding them, discarding information orthogonal to the leading principal components; concatenation retains 64 independent embedding dimensions alongside all 60 raw features, preserving representational capacity. Summation variants ($C = 0.584$–0.596) fall below the cross-sectional CoxPH baseline (0.619), while concatenation variants (0.622–0.624) exceed it, making fusion strategy a more consequential choice than encoder selection. The Setting 4 advantage over Setting 3 is preserved under concatenation but compressed under summation (BCB·sum: 0.596 vs. 0.584), consistent with information loss from the dimensionality reduction step.

Calibration gap of autoregressive generation. Delphi's IBS of 0.185 vs. $\approx$0.141 for supervised CoxPH variants must be interpreted alongside an important asymmetry: Delphi operates solely on serialised ICD codes and demographic tokens and has no access to the numeric biomarker features (HbA1c, CRP, creatinine, cystatin C) available to the CoxPH baselines. This input asymmetry means the gap partly reflects missing acute biomarker signal rather than a deficiency of the autoregressive approach within its intended operating regime. More fundamentally, the sigmoid hazard chain produces survival curves without a calibration reference from the target population, an inherent constraint of zero-shot deployment under distribution shift between pre-training and the UK Biobank cohort. The near-term AUC collapse (0.52 at 9.5 yr) is consistent with this interpretation: short-horizon mortality risk is dominated by derangements in CRP, creatinine, and albumin that do not appear in the ICD token stream. These results therefore reflect a structural limitation of the autoregressive generation paradigm for *acute* risk stratification: the paradigm lacks access to the quantitative physiological signal that most strongly predicts near-term death. Fine-tuning with a calibration-aware objective and numeric biomarker inputs would likely close this gap substantially, but at the cost of zero-shot generalisability across cohorts.

Limitations. All embedding methods freeze their encoders; fine-tuning may yield additional gains at substantially higher compute cost. IBS and TD-AUC rely on independent censoring assumptions that may be violated in UKB. All results are from a single biobank cohort; external validation is required before clinical translation. Bio_ClinicalBERT's 512-token limit is an architectural constraint that disproportionately affects Setting 5; future work should evaluate longer-context clinical encoders. The primary evaluation metrics (C-index, TD-AUC) measure discrimination but not calibration of the predicted survival function. We supplement IBS with restricted mean survival time ($\mathrm{RMST} = \int_0^\tau S(t \mid x)\,\mathrm{d}t$), which provides a patient-interpretable calibration summary (expected life-years within the follow-up window τ). RMST-decile calibration analysis across the full 12-cell matrix is left as an extension once all server embedding runs are consolidated locally.

6 Conclusion

We have presented a controlled three-axis study of LLM-based and classical survival prediction for all-cause mortality on a UK Biobank respiratory disease subgroup (n = 124,158). By varying input representation (5 settings), encoder architecture (2 encoders), and feature fusion method (2 fusion strategies) under a uniform CoxPH survival head, we isolate which structural encoding choices drive prediction quality.

Three findings stand out. First, *temporal ordering is the primary discriminative structural property*: decoupled age–event encoding (Setting 4) outperforms prose serialisation of identical events by +0.008 C-index, regardless of encoder or fusion. Second, *general-purpose LLMs implicitly encode numeric biomarker structure*: a Qwen3-8B prose embedding without any raw numeric features matches a fully supervised 39-feature CoxPH, with superior TD-AUC. Third, *autoregressive generation has a structural calibration gap*: Delphi's near-term AUC collapse and high Brier score reveal that pre-trained EHR LLMs without task-specific grounding are insufficient for calibrated risk stratification.

Together, these findings confirm that *how* temporal structure is encoded matters more than which specific LLM encodes it, corroborated by both the encoder ablation (Bio_ClinicalBERT matches Qwen3-8B on Settings 3 and 4) and the fusion ablation (concatenation consistently outperforms summation by +0.028–+0.039 C-index). Future work will explore fine-tuning with calibration-aware objectives, longer-context clinical encoders, and multimodal fusion of EHR embeddings with imaging and genomic modalities.

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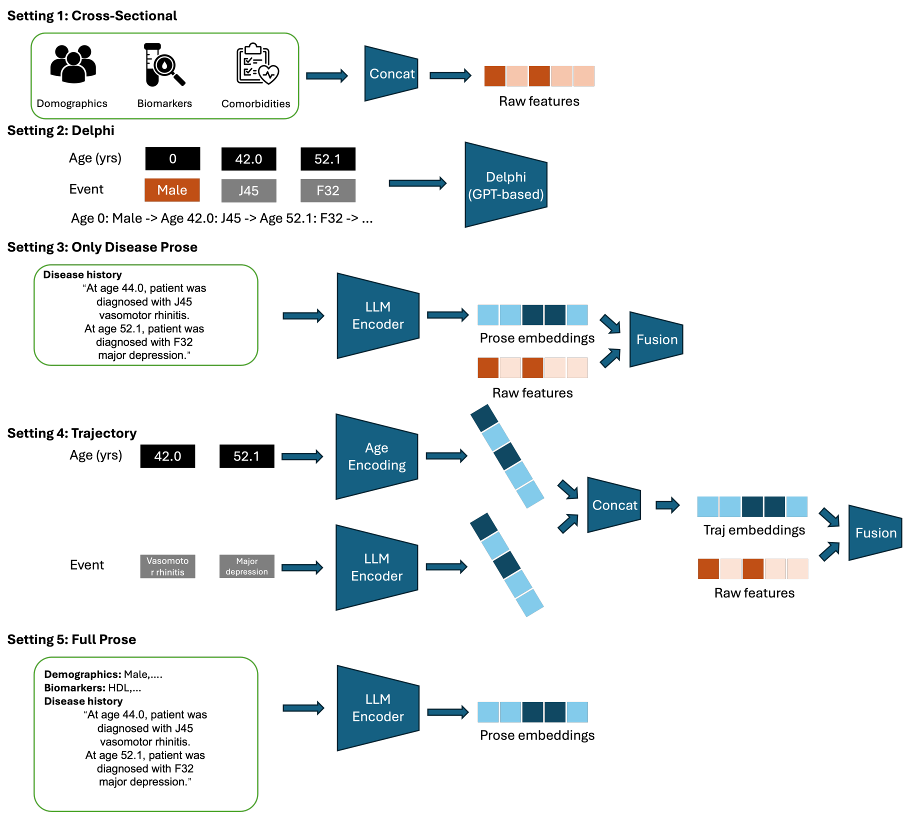

Fig. 1. Overview of the five experimental settings and their encoding pipelines. **S1** (top): demographics, biomarkers, and comorbidities are concatenated into a fixed-dimension feature vector for CoxPH. **S2**: age-tagged ICD tokens are fed to a GPT-based generative transformer (Delphi) in zero-shot mode. **S3**: disease history is serialised as natural-language prose and encoded by a frozen LLM; the resulting embedding is fused with raw features. **S4**: age values are mapped to a continuous sinusoidal encoding in parallel with ICD token embeddings; the two streams are concatenated per event, mean-pooled, and fused with raw features. **S5** (bottom): all modalities are combined into a single paragraph and encoded by the LLM with no raw features concatenated.

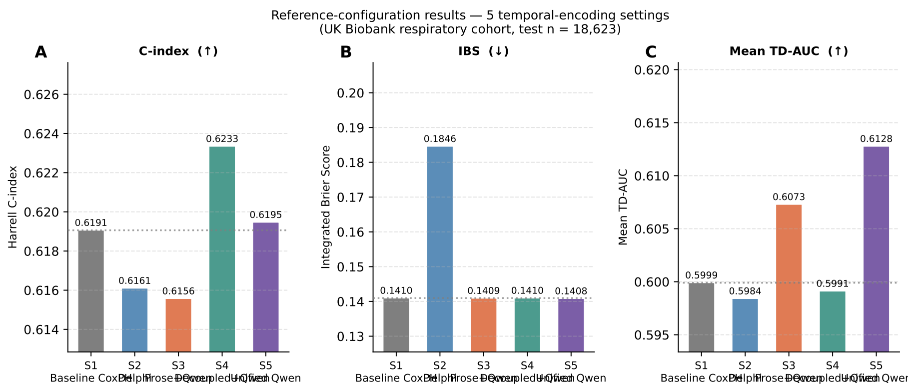

Fig. 2. Reference-configuration results for the five input-format settings. (A) Harrell C-index; (B) Integrated Brier Score (lower is better); (C) Mean TD-AUC. The dotted line marks the S1 CoxPH baseline value in each panel. All embedding settings use Qwen3-8B with concatenation fusion where applicable.

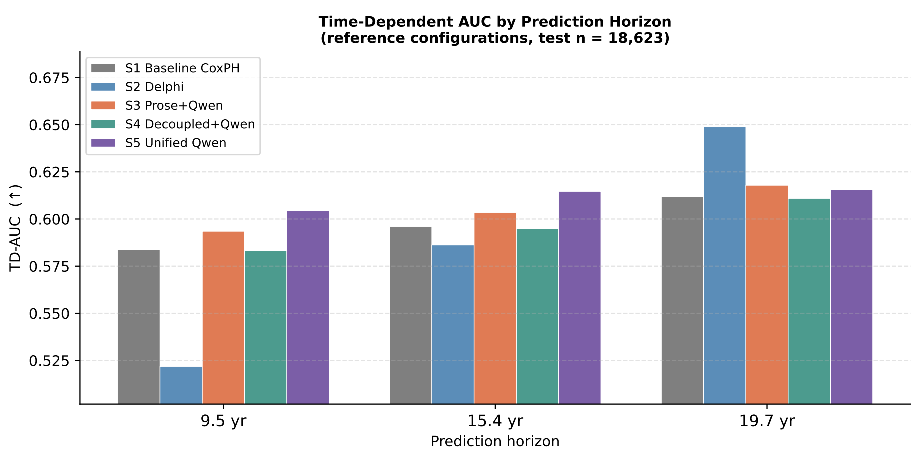

Fig. 3. TD-AUC by prediction horizon for the five reference configurations. Delphi (S2) rises from near-chance (0.52) at 9.5 yr to peak (0.65) at 19.7 yr, while CoxPH-based settings show a comparatively flat profile across horizons.